CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 23: Trigonometrical Ratios of Standard Angles

Every formula and result of the chapter on one sheet

1	$\sin 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$	$0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$
2	$\cos 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$	$1, \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2}, 0$
3	$\tan 0^\circ, 30^\circ, 45^\circ, 60^\circ$	$0, \frac{1}{\sqrt{3}}, 1, \sqrt{3}$
4	$\cot 30^\circ, 45^\circ, 60^\circ, 90^\circ$	$\sqrt{3}, 1, \frac{1}{\sqrt{3}}, 0$
5	$\sec 0^\circ, 30^\circ, 45^\circ, 60^\circ$	$1, \frac{2}{\sqrt{3}}, \sqrt{2}, 2$
6	$\operatorname{cosec} 30^\circ, 45^\circ, 60^\circ, 90^\circ$	$2, \sqrt{2}, \frac{2}{\sqrt{3}}, 1$
7	Not defined	$\tan 90^\circ, \sec 90^\circ, \cot 0^\circ, \operatorname{cosec} 0^\circ$
8	30-60-90 triangle	Sides $1 : \sqrt{3} : 2$
9	45-45-90 triangle	Sides $1 : 1 : \sqrt{2}$
10	Double angle (to verify)	$\sin 2A = 2 \sin A \cos A$
11	Double angle (to verify)	$\cos 2A = \cos^2 A - \sin^2 A$
12	Equal sine and cosine	$\theta = 45^\circ$